MULTI-AGENT SYSTEM

Agentic Sentiment Analysis

Babysitter Churn Detection Workflow

Defining agents, tools, and open-source LLMs for chat-based sitter retention intelligence

System Architecture

5-Agent CrewAI Crew — Chat Database → Churn Risk → Outreach

Supervisor — Orchestrator → **Data Collector** — Ingestion → **Perception Agent** — Surface signals → **Reasoning Agent** — Deep context → **Reporter / Action** — Output + outreach

Chat DB input

Outreach sent

01 Supervisor triggers rolling pipeline per sitter from chat DB

02 Data Collector fetches, cleans, and chunks conversation history

03 Perception + Reasoning run in parallel on same preprocessed data

04 Reporter synthesizes outputs and dispatches personalized outreach

Agent 1 — Supervisor / Orchestrator

Churn Watch Orchestrator

Role

Central coordinator and risk profile manager for all active sitters on the platform. Routes inputs to specialist agents, aggregates outputs, and controls escalation logic.

Goal

Trigger the pipeline for each sitter on a rolling 24-hour basis. Aggregate specialist outputs into a unified churn risk score. Ensure no at-risk sitter goes unaddressed.

Backstory

A seasoned platform ops lead who's seen hundreds of sitters go silent. Knows one signal is noise — a cluster is a pattern. Never acts on a single agent's output alone.

Tools

- chat_db_connector
- sitter_profile_fetcher
- risk_score_aggregator
- pipeline_scheduler
- escalation_router

Agent 2 — Data Collection

Chat Preprocessing Agent

Role

Raw chat data cleaner and structurer. Pulls sitter conversation history from the database, segments by type (sitter↔parent, sitter↔support), and passes clean data downstream.

Goal

Strip PII, chunk into time windows (7/14/30 days), preserve timestamps and thread IDs. Ensure no dirty data reaches specialist agents — quality gate for the whole pipeline.

Backstory

A meticulous data engineer who refuses to let bad data corrupt good analysis. Knows that 'k' at 11pm is context-dependent — so it preserves metadata alongside the text.

Tools

- SQLAlchemy / psycopg2
- PII_anonymizer
- message_chunker
- sitter_convo_splitter *
- time_window_batcher *

* custom-built tool

Agent 3 — Perception

Tone & Surface Signal Agent

Role

Surface-level linguistic analyst. Reads lexical sentiment signals — polarity words, intensifiers, negations, emoji tone — and emits a preliminary label with a confidence score.

Goal

Score sentiment per message and per time window. Detect tone drift (positive → neutral → negative trajectory). Flag sitters whose language shifted from warm to clipped or complaint-heavy.

Backstory

A linguist specializing in affective vocabulary. Reads fast, flags ambiguous tokens, tracks the trajectory not just the snapshot. Knows 'ok confirmed' after 'I had SO much fun!' is a signal.

Tools

- VADER scorer
- HuggingFace sentiment pipeline
- emoji_tone_interpreter
- tone_trajectory_scorer *
- warmth_index_calc *

* custom-built tool

Agent 4 — Reasoning

Deep Context & Pragmatics Agent

Role

Deep context and pragmatics interpreter. Resolves sarcasm, irony, mixed-emotion sentences, and domain-specific idioms that surface classifiers miss entirely.

Goal

Use RAG over the sitter's full conversation history to surface slow-burn dissatisfaction. Understand thread context, detect complaint patterns, catch what Perception cannot — the subtext.

Backstory

A pragmatics expert who knows 'great, another last-minute cancel' is not enthusiasm. Leverages conversation thread context and domain KB via RAG. Reads what the sitter means, not just what they wrote.

Tools

- LangChain RAG pipeline
- FAISS vector index
- sentence-transformers
- sarcasm_classifier *
- ghost_event_detector *

* custom-built tool

Agent 5 — Output & Action

Reporter / Retention Outreach Agent

Role

Structured output formatter, churn risk synthesizer, and downstream action trigger. Combines Perception and Reasoning outputs into one verdict, then acts on it.

Goal

Produce a churn risk level (Low / Medium / High / Critical) with a plain-English rationale. Draft a personalized outreach message and recommend one concrete retention action matched to root cause.

Backstory

A community manager who knows a sitter frustrated about late-cancels doesn't want a generic 'we value you' email. Matches tone and action to root cause. Never sends the same template twice.

Tools

- Jinja2 templating
- SMTP / Twilio
- risk_level_classifier
- root_cause_to_action *
- outreach_history_check *

* custom-built tool

Tools Overview

Standard libraries vs. custom tools built for this workflow

SQLAlchemy / psycopg2
Standard

Agent: Data Collector

Query chat DB by sitter ID, date range. Returns structured message JSON.

VADER + HuggingFace pipeline
Standard

Agent: Perception

Polarity scoring on short chat text. cardiffnlp/twitter-roberta for social tone.

LangChain RAG (FAISS)
Standard

Agent: Reasoning

Embed conversation history, query semantically for slow-burn frustration signals.

Jinja2 + SMTP / Twilio
Standard

Agent: Reporter

Template-fill personalized outreach messages, dispatch via email or SMS.

sitter_convo_splitter
Custom

Agent: Data Collector

Separates parent↔sitter vs. support↔sitter threads per sitter ID.

tone_trajectory_scorer
Custom

Agent: Perception

Rolling sentiment delta across 7/14/30-day windows vs. sitter baseline.

ghost_event_detector
Custom

Agent: Reasoning

Flags mid-thread ghosting — conversations where sitter stopped replying.

root_cause_to_action_mapper
Custom

Agent: Reporter

Maps complaint category → retention lever (pay review, parent flag, recognition).

Open-Source LLMs

Best models for sarcasm, vague conversational language, and psychological subtext

Best Overall

Llama 3.1 70B Instruct

by Meta

Strongest open-source model for nuanced conversational reasoning. Excels at sarcasm detection, tone inference, and following complex instructions. Use for Reasoning Agent and Reporter.

KEY STRENGTHS

- Sarcasm & irony resolution
- Long-context chat threads
- Rich rationale generation

Assign to: Reasoning · Reporter

Fast Classifier

Mistral 7B / Mixtral 8x7B

by Mistral AI

Extremely fast inference with strong classification capability. Mixtral's MoE architecture handles high-throughput sentiment tagging efficiently at low cost.

KEY STRENGTHS

- High-throughput classification
- Structured output reliability
- Low cost per token

Assign to: Perception · Data Collector

Conversational + Multilingual

Qwen2.5 14B Instruct

by Alibaba

Exceptional at conversational text analysis and real chat data. Strong psychological inference and multilingual support. Best choice if sitter base includes non-English speakers.

KEY STRENGTHS

- Real chat language comprehension
- Multilingual sentiment
- Psychological subtext reading

Assign to: Perception · Reasoning

One LLM or Many?

You don't need multiple LLMs — you need multiple agents. Here's the decision framework.

An LLM is the brain. An agent is a role given to that brain with a specific system prompt, tool access, and memory scope. You can run a 5-agent crew off a single model.

Start here: One LLM

- Llama 3.1 8B handles the whole crew on a single GPU
- Each agent = different system prompt, same model
- Simple to debug — one model, five contexts
- Cheap to run — no multi-model inference cost
- Best starting point for any new deployment

Optimize later: Mix models

- Swap Mistral 7B into Perception for 5x faster classification
- Upgrade Reasoning to Llama 70B for sarcasm accuracy
- Use Qwen2.5 if sitters message in multiple languages
- Profile first — only swap where it moves the needle
- CrewAI supports per-agent llm= parameter natively

System Summary

What you're building

5 — **Specialized Agents**

Supervisor, Collector, Perception, Reasoning, Reporter

4 — **Core Tools**

SQLAlchemy · VADER/HuggingFace · LangChain RAG · Jinja2+SMTP

4 — **Custom Tools**

convo_splitter · tone_trajectory · ghost_detector · action_mapper

1 — **LLM to start**

Llama 3.1 8B — one model, five agents, full crew

The Conflict Resolver agent is ready to plug in between Reasoning and Reporter when needed.